

Statistical Estimation – Complete Study Notes

1. Statistical Estimation

Statistical Estimation is the process of using observed sample data to estimate unknown values (parameters) of an entire population.

2. Population vs Sample

Population: Complete collection of observations.

Sample: A subset selected from the population to draw conclusions.

3. Parameter vs Statistic

Parameter: Numerical measure of a population (μ , σ , p).

Statistic: Numerical measure computed from a sample ($\bar{x}$, s , p).

4. Estimator vs Estimate

Estimator: A mathematical rule or formula used to estimate a parameter.

Estimate: The numerical value obtained after applying the estimator to sample data.

5. Point Estimation

Provides a single best value for an unknown parameter.

6. Interval Estimation

Provides a range of likely values (Confidence Interval) instead of one value.

7. Good Estimator Properties

- Unbiasedness
- Consistency
- Efficiency (minimum variance)
- Sufficiency
- Stability (low variance).

8. Training vs Testing

Training data is used to build the model. Testing data checks how well it generalizes to unseen data.

9. Bias

Bias is a systematic error caused by overly simple assumptions. High bias generally causes underfitting.

10. Variance

Variance measures how much model predictions change when trained on different datasets. High variance causes overfitting.

11. Bias–Variance Tradeoff

Simple Model → High Bias, Low Variance → Underfitting.

Complex Model → Low Bias, High Variance → Overfitting.

Best models balance both.

12. Underfitting

Model is too simple and fails to learn important patterns.

13. Overfitting

Model memorizes training data, performs poorly on unseen data.

14. Acronyms

MAE – Mean Absolute Error.

MSE – Mean Squared Error.
RMSE – Root Mean Squared Error.
MAPE – Mean Absolute Percentage Error.
MLE – Maximum Likelihood Estimation.
MAP – Maximum A Posteriori.

15. MLE vs MAP

MLE estimates parameters using only observed data. MAP combines observed data with prior knowledge using Bayes' theorem.

16. Standard Error (SE)

$SE = \sigma/\sqrt{n}$. Measures variability of sample means.

17. Common Interview Questions

Difference between estimator and estimate?
Difference between parameter and statistic?
Why is bias-variance tradeoff important?
Difference between MLE and MAP?
Explain underfitting vs overfitting.

18. Key Takeaways

Estimator = Formula, Estimate = Value.
High Bias → Underfitting.
High Variance → Overfitting.
Goal = Low Bias + Low Variance.
Training accuracy alone is not enough; good test performance indicates generalization.